Meet Babariya

AI Engineer · LLM systems, evaluation & ML benchmarks · quantitative-finance background

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Summary

Business Information Systems student (grade 1.55) who builds and evaluates AI systems. At **Datacurve (Y Combinator)** I design ML benchmark challenges used to train and test autonomous AI agents. I have built RAG, tool-using agents, vision fine-tuning pipelines and a full-stack AI trading product. Kaggle Competitions Expert (top 1%). Seeking an **AI / ML engineering internship or Werkstudent role** in Germany.

Experience

Datacurve (Y Combinator) — Project Eris Remote

ML Benchmark Engineer (Contract) *Sep 2026 – Present*

- Design end-to-end ML benchmark challenges used to train and evaluate AI coding agents: dataset curation, split pipeline (*prepare.py*), evaluator (*grade.py*), scoring rubrics and reference solution notebooks.
- Built 25+ challenge designs across **vision fine-tuning, sequence-to-sequence, from-scratch and LLM-evaluation** tracks, sourcing commercially licensed real-world data (Zenodo, NOAA, Hugging Face).
- Calibrated difficulty so frontier agents score below 0.8: baseline ladders from frozen-feature ridge floors to GPU ResNet fine-tunes, plus exact Bayes-optimal decoders (forward–backward) that prove a task's information ceiling.
- Hardened graders against label leakage, test-set lookup and malformed submissions; designs cleared automated novelty, leakage, evaluator-contract and agent-calibration gates.

Jainam Broking Surat, India

Derivatives Trader *Apr 2022 – Dec 2024*

- Traded weekly NIFTY / BANKNIFTY options with delta-neutral strategies and real-time, Greeks-based hedging.

Education

Furtwangen University (HFU) Furtwangen, Germany

B.Sc. International Business Information Systems *Mar 2025 – Present*

- Current grade: **1.55** · Transcript

Projects

QuantPilot — AI trading copilot & quant-ranked copy trading | *TypeScript, Next.js, Hono, Python, FastAPI, PostgreSQL* *code private*

- Full-stack product on Hyperliquid: TypeScript execution API, Telegram bot and Next.js signing app over a Python quant service with a shared PostgreSQL schema. Non-custodial agent-wallet trading is verified on testnet.
- Ranks traders by risk- and funding-adjusted returns with a Deflated-Sharpe filter against luck. The ranking worker has run on live mainnet data.

Financial Filings RAG | GitHub | *Python, Claude API, sentence-transformers, ChromaDB*

- Question answering over SEC 10-K/10-Q filings: chunking, local embeddings and vector search, with a grounded, citation-backed prompt that limits answers to the retrieved context so the model can't invent financial figures.

Financial Analyst Agent | GitHub | *Python, Claude tool use, yfinance*

- Autonomous agent with a hand-written tool-use loop that chains market-data and risk tools (VaR, volatility, Sharpe) before writing a sourced analysis.

German Traffic Sign Recognition | GitHub | *PyTorch, CNN, Streamlit*

- CNN for 43-class GTSRB (ADAS perception) reaching **97.7% test accuracy** with 676K parameters. Covers augmentation, LR scheduling, evaluation and a Streamlit demo.

Competitions & Achievements

- **Kaggle Competitions Expert** (top 1% globally): Silver, CIBMTR (120th / 3,325); Bronze, MITSUI Commodity Prediction (165th / 1,711).
- **IMC Prosperity 4** (algorithmic trading): 230th / 18,803 teams worldwide, **12th in Germany**.
- **WorldQuant BRAIN**: Gold level · **AlgoTrade Hackathon 2025** (Zagreb) · Applied Data Science Lab, WorldQuant University.

Technical Skills

LLM / GenAI	Claude & OpenAI APIs, RAG, embeddings, vector DBs (ChromaDB), agents & tool use, LLM evaluation, benchmark & rubric design, prompt engineering
ML / Deep Learning	PyTorch, Hugging Face, fine-tuning (vision backbones), scikit-learn, gradient boosting, CNNs, time-series modelling
Software	Python, TypeScript, SQL, C++; Next.js / React, FastAPI, PostgreSQL, Docker, Git; pandas, NumPy, Spark, Power BI
Languages	English (fluent), German (A2, actively learning)